Analytical Chemistry Cumulative Exam
March 29, 2013

Basic Terminology, Theory, and Techniques of Electroanalytical Chemistry

11. (36 points) Define the following terms in the context of electrochemistry. Draw diagrams or schematics to assist with your definitions wherever needed:

(a) Double layer.

(b) Inner Helmholtz plane.

(c) Outer Helmholtz plane.

(d) Diffuse Layer.

(e) Diffusion Layer.

(f) Transference number.

(g) The transfer coefficient.

(h) Faradaic current.

(i) Potentiostat.

(j) Galvanostat.

(k) Cyclic voltammetry

(l) Chronoamperometry

2. (21 points) Compare the following features at microelectrodes to those at macroelectrodes. (Assume that both of the electrodes are made of the same material and both have the same shape (such as disk shape), and both are placed in the same solution that contains both a supporting electrolyte and an electroactive species that can be oxidized or reduced at the potentials applied for the experiment.)

(a) Double layer capacitance (the absolute value).

(b) Double layer capacitance that is normalized to the electrode area.

(c) Faradaic current that is normalized to the electrode area (this is called faradaic current density).

(d) The ratio of the faradaic current to the charging current.

(e) Performance at fast experimental time scales.

(f) Performance in solutions where there is a high uncompensated resistance.

(g) Effect on altering the composition of the redox species in solution during an electrochemistry experiment.

3. (10 points) What is a reference electrode? Describe the features of an ideal reference electrode.

4. (10 points) In a primary lithium battery, the reaction that takes place at the anode when the battery discharges is:

$$Li = Li^{+} + e^{-}$$

The standard electrode potential for the *reduction* is -3.045 V vs. NHE at 25 °C. What is the standard electrode potential for Li^{+} reduction to Li vs. the SCE reference electrode at 25 °C? (Use the diagram at the right to assist you in this calculation.)

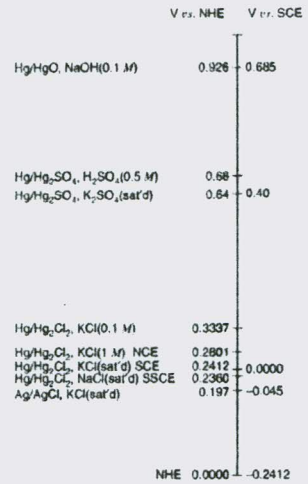
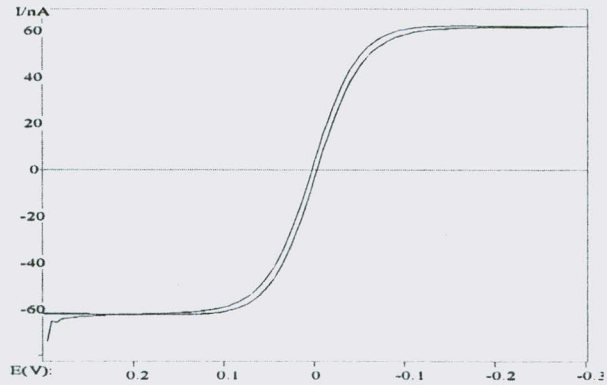


Figure E.1 Potentials of reference electrodes in aqueous solutions at 25°C. [See D. J. G. Ives and G. J. Janz, "Reference Electrodes," Academic, New York, 1961.]

5. (13 points) Take a look at the following figure, which is a cyclic voltammogram at an electrode for redox species in a semi-infinite, static solution that involves a fast, reversible, one-electron transfer, with stable and soluble products and reactants (no following chemistry or preceding chemistry). Based on the data, comment on the *size* of the electrode *compared* to the *thickness* of the diffusion layer and the *relative concentrations* of the oxidized and reduced forms of the redox couple. Also report the *standard electrode potential* that you can obtain, if you assume that the temperature is 25 °C, the diffusion coefficients of the both forms of the redox couple are equal and the activity coefficients of those forms are also equal?



6. (10 points) An array of coplanar platinum microdisks inlaid in an insulator and shorted together is used as an electrode. (The diameter of each disk is 10 μm. The microdisks are placed in a regular x-y-grid arrangement with a 100-μm center-to-center spacing (leaving a 90-μm insulator gap between the edges of adjacent disks). The *reduced* form of a soluble and kinetically reversible redox couple is present in solution. What would the cyclic voltammetry response look like under conditions where $(D\ t)^{1/2}$ is *larger* than the separation between the microdisks? (D is the diffusion coefficient and t is the time of the experiment.)